

41 — Bench Bill of Materials

v1.0 — what to buy, in what order, to what spec

Function: Turn the test programs of docs 12 §4 (liquid, tests A–L) and docs 21–22 (solid, T1–T9 / M1–M4) into a procurement list a stranger can act on without re-deriving the author's choices. The documents say *what each experiment decides*; this says *what to put in the basket, in what quantity, and which parameter of each item is acceptance-critical*. It adds no experiment, changes no figure, and decides nothing — where it and a track document disagree, the track document wins.

1. How to read this list — and what it deliberately omits

No part numbers, no vendors, no prices. Every entry is a *class plus the spec that matters*. This is deliberate and it is not laziness:

- A SKU is the fastest-rotting thing in a record meant to be read decades from now — faster than a price, which doc 12 §4 already scopes as "2026 order-of-magnitude, one currency, one region."
- Naming a vendor implies a validation that has not happened. Nothing here is built (doc 00 §9); no item on this list has been bought, wired, or run.
- What actually transfers is the **specification**: "±2% RH over 20–90%, with a calibration certificate" survives translation to any market and any decade. "Model X" does not.

Where a spec is genuinely load-bearing — the case where a cheaper part silently invalidates the result — it is called out as **acceptance-critical**.

Confidence grades follow doc 00 §9. *Procurement-grade* means the item and its spec are ordinary commerce and can be sourced today. *Sizing-grade* means the **quantity** derives from a design figure that is itself graded, so it moves if that figure moves. Nothing here is *measured*.

The shared instrumentation spine is specified in doc 22 §7 and is not restated here — including the reason wet/dry-bulb psychrometry, not a capacitive sensor, is required at the near-saturated and hot-regen nodes, and the channel-budget warning. Buy that stack once; every test below reuses it. This document lists only what each test adds on top.

1.1 What this list is not

There is no system bill of materials, and there should not be one yet. Plate thickness, channel pitch, coated area and per-module fluid inventory are all open — that is finding **F6**, gated on **M1** and **M3** — and the effective Δq that sets sorbent inventory is **PENDING M2 (F3)**. A parts list for the built system would therefore be a list of ungraded numbers wearing part numbers, which is precisely what doc 00 §9 forbids. The bench program exists to close those gates. This list buys the bench, not the machine.

2. Stage 1 — the cheap parallel set

Everything here runs concurrently and none of it gates anything else. Per doc 12 §4 and the README reconstruction path, this is where a reconstructor starts.

2.1 Test A — jar equilibrium (*the single highest-value \$30 in the program*)

Decides the real a_w table that feeds every floor number and the X12 lift inequality. Go/no-go: **40 wt% @ 33 °C ≤ 55% ERH**.

Item	Spec that matters	Qty	Grade
Wide-mouth jars, sealing lid	PP or glass; lid must take a sensor pass-through without leaking; ~250–500 mL	6–10	procurement
Capacitive RH/T sensor	Acceptance-critical: ±2% RH or better over 20–90% RH, with a calibration certificate. SHT85/ SHT45 class per doc 22 §7	1–2	procurement
Calcium chloride	Food or technical grade, anhydrous or dihydrate; state which — the hydrate changes the mass you weigh out	~1 kg	sizing
Lithium chloride	Comparison arm only; small quantity	~50 g	sizing
Balance	0.01 g resolution at ≥ 200 g capacity	1	procurement
Stable warm enclosure	Must hold 27 °C and 33 °C setpoints; an incubator, proofing box or sous-vide bath — the same warm stack tests D/G/L reuse	1	procurement
Saturated-salt calibration standards	For checking the RH probe before trusting it: LiCl, MgCl ₂ , NaCl class	1 set	procurement

Quantities are sizing-grade: they assume ~100–200 mL of brine per concentration across the 35/38/40/42/44 wt% ladder plus the LiCl arm, with enough spare to remake a jar.

The one place not to economise: the RH probe. The go/no-go sits at 55% ERH and the interesting band is 45–60%; a $\pm 5\%$ sensor cannot resolve a pass from a fail.

2.2 Test B — aerosol drift (~\$5, and it gates cabin connection)

Decides whether a liquid contactor may ever touch breathing air (X3), and feeds the doc 00 §8 safety register item 3.

Item	Spec that matters	Qty	Grade
Bare mild-steel coupons	Acceptance-critical: bare and unpainted — the whole assay is that rust is visible. Any coating defeats it	4–6	procurement
Drift eliminator / demister media	The chain being tested, per doc 11 §3	as designed	sizing
Small fan and duct section	Must reach the maximum design face velocity , not a comfortable one	1	sizing
Brine	Shares test A's CaCl_2	—	—

Run both **trickle and flooded**. The flooded case is the one that fails.

2.3 Test E — ERV core

Decides ϵ_{lat} , the fouling trend, and **CO₂ crossover < 5%** (X7, and the interlock's credibility).

Item	Spec that matters	Qty	Grade
Enthalpy-recovery core	Membrane type, stated ϵ ; sized to the doc 00 §5 fresh-air flow	1	procurement
NDIR CO ₂ sensors	Acceptance-critical: two matched units — crossover is a <i>difference</i> measurement, so an unmatched pair reports the offset, not the leak	2	procurement
U-tube manometer or differential pressure gauge	For the ΔP fouling trend	1	procurement
Salt-aerosol source	To drive the fouling trend at accelerated rate	1	sizing

2.4 Tests J and J-K — CO₂ sorbents (required-PENDING; gates X10/X11)

The **breathing-air gate**: amine/ammonia slip for J, alkaline particulate/mist carryover for J-K (doc 00 §8 item 4).

Item	Spec that matters	Qty	Grade
Solid amine sorbent	Lewatit VP OC 1065 class	small lot	procurement
K ₂ CO ₃ on apolar carbon	J-K only; acceptance-critical: apolar carbon or TiO₂ support — alumina and MgO are prohibited (X11), and an alumina control is part of the test	small lot	procurement
NDIR CO ₂ sensor, ppm-resolving	Must resolve 1,000–1,500 ppm , not just percent-level	1	procurement
Regeneration heat source	J: 85–95 °C. J-K: 120/135/150 °C setpoints — the higher band needs a different heater than the sous-vide stack	1	procurement
Slip assay	Amine/ammonia detection tubes or equivalent; alkaline carryover by wipe/filter	1 set	procurement

Run J-K only where a $\geq \sim 130$ °C tap exists on the platform (X11).

2.5 Solid track T1 and T9 — feedstock and benchmark

Item	Spec that matters	Qty	Grade
Aluminium hydroxide feedstock	Acceptance-critical to F4: record whether it is freshly precipitated amorphous Al(OH)₃ or crystalline gibbsite. A PXRD fail may be the feedstock, not the milling	per doc 21 §4	sizing
Fumaric acid	Technical grade	per doc 21 §4	sizing
Mortar and pestle	The mill-zero route (T1)	1	procurement
Commercial AlFu lot	Basolite A520 class, for the T9 benchmark arm	small lot	procurement
Outsourced PXRD + DVS	A service, not a purchase — book it before synthesising	per T4	procurement
T8 silica-wheel benchmark quote	Also a solicitation, not a purchase: request it in parallel at zero cost — solid-track hardware waits on it as well as on M2 (doc 21 §2)	1 quote	procurement

3. Stage 2 — test I, the liquid gate

Nothing downstream on the liquid track is worth buying until test I reports. Measure **irrigation rate first** — it is the variable doc 12 §4 names as the first-order one, and CHK-019 records that the cell rating, face-velocity band and NTU approach do not currently share an operating point.

Item	Spec that matters	Qty	Grade
Film-cell structured packing	Per doc 11 §2; ~600 × 300 × 150 mm class	1–2 cells	sizing
Wetted-path materials	Acceptance-critical: the doc 11 §1 two-worlds law is absolute. PP / PVC / PVDF / PE-RT / titanium only. No copper, brass, aluminium, zinc, 304 SS, and no nylon — CaCl ₂ stress-cracks loaded polyamide	all	procurement
Wetted fasteners	PP / PVC / PVDF only. Plumbing-aisle parts hide brass	all	procurement
Distribution and pump	Must give a controllable, measurable irrigation rate — the primary variable	1	sizing
Rocking rig	5–20° for the marine case; also the 30° passive-path rating (IN-016)	1	sizing
Flow measurement	Timed volumetric catch is sufficient (doc 22 §7)	—	—

Never a brazed-plate exchanger anywhere brine or raw water flows: 304 plus copper braze is a two-worlds violation (doc 11 §1).

4. Stage 3 and 4 — after the gates

Deliberately thin. Tests D, G and H, and the M1–M4 module set, reuse the warm stack, the psychrometric stack and the fluid loop already bought above; doc 22 §5 specifies the single-coupon rig and doc 22 §7 the channels. The additions are:

Test	Adds	Grade
D — regeneration COP	Still tray; condenser with a measurable approach	sizing
G — sealed-still rate	TDS meter — potability and entrainment both hinge on it (doc 00 §8 item 5)	procurement
H — M-cycle wetting	Wicking media; tilt fixture 5/10/15°; closed-loop feed path for the X8 point	sizing
M1–M4	Second logger or the unified ESP32 + MAX31856/SHT node — see the doc 22 §7 channel-budget warning; gravimetric balance; oven with independently thermocouple-verified setpoints	procurement
F — endurance	Nothing. It is a locker and patience	—

5. Reagent and consumable summary

Consumable	Feeds	Indicative quantity	Grade
Calcium chloride	A, B, I, and the X12 arm	~1 kg for A and B; I is the step change and its charge follows the cell design	sizing
Lithium chloride	A comparison arm	~50 g	sizing
Fumaric acid + Al(OH) ₃	T1, T4	per doc 21 §4	sizing
Solid amine / K ₂ CO ₃ -on-carbon	J, J-K	small lots	procurement
Mild-steel coupons	B, and any materials re-check	4–6	procurement
Saturated-salt RH standards	Probe calibration before A	1 set	procurement
Distilled water	The whole solid-track loop; M-cycle feed	ongoing	procurement

6. Sequencing note

The order above is the README's cheapest-decisive-first path and it is not arbitrary: **A** moves the largest single unknown for ~30, * * B * *
*canvetotheentireliquidcabin – airarchitecturefor 5, and **I** is the gate that decides whether any liquid hardware is worth building. Buying stage 2 before stage 1 reports is how a program spends its budget on the wrong answer.

Part of an open defensive-publication release: hardware CERN-OHL-P v2, text CC-BY-4.0, scripts MIT. No patents sought or held. Unbuilt paper design — see LICENSE for the safety disclaimer.

Version history

- **v1.0** — New document. Consolidates the procurement layer of the doc 12 §4 and doc 21–22 test programs into one ordered list, specified by class and acceptance-critical parameter rather than by part number. Records explicitly that no system bill of materials exists or should yet exist, because the geometry it would enumerate is open per F6/F3 and gated on M1/M2/M3. No experiment added, no figure changed.